Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 34: Approved Solutions

Permutations

Covers Chartrand & Zhang, Section 8.4

Status of this set

Ungraded practice. Solutions released with it. Show the expression, not just the number.

Part I: How to Practice This Material

The formula and where it comes from

$$P(n,r) = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}$$

This is the Multiplication Principle with one fewer choice at each stage. If you forget the formula, rebuild it: n choices, then $n-1$, and so on for r stages.

The test

Does reordering the chosen items change the outcome? If yes, it is a permutation.

Three mistakes to expect

1. **Using $P(n,r)$ when order does not matter.** That over-counts by $r!$.
2. **Ignoring a restriction** — a leading digit, a fixed position.
3. **Giving a bare number.** The expression is the reasoning.

Part II: Practice Problems

Section A — Direct counts

1. Compute each, showing the product.

(a) $P(9,3)$ (b) $P(6,6)$ (c) $P(12,2)$

Approved solution

(a) $9 \cdot 8 \cdot 7 = 504$. **(b)** $6! = 720$. **(c)** $12 \cdot 11 = 132$.
Note (b): $P(n, n) = n!/0! = n!$, which works only because $0! = 1$.

2. In how many ways can 7 distinct books be arranged on a shelf? In how many ways if two specific books must be adjacent?

Approved solution

All arrangements: $7! = 5040$.
Two adjacent: glue the pair into a single block. Now arrange 6 objects: $6! = 720$. The two glued books can be in either internal order, so multiply by 2: $720 \cdot 2 = 1440$.

The gluing trick handles every "must be together" condition: treat the group as one item, arrange, then multiply by the arrangements *within* the group.

Section B — Restrictions

3. How many four-letter strings can be formed from the letters of NUMBERS with no repeated letter? How many begin with a vowel?

Approved solution

NUMBERS has 7 distinct letters.
All: $P(7, 4) = 7 \cdot 6 \cdot 5 \cdot 4 = 840$.
Beginning with a vowel: the vowels are U and E, so 2 choices for the first letter, then $P(6, 3) = 6 \cdot 5 \cdot 4 = 120$ for the rest: $2 \cdot 120 = 240$.

Handle the restricted position first. Choosing the first letter before the others keeps the remaining counts fixed — the Lecture 31 lesson about interfering stages.

4. A club of 10 members elects a president, a vice-president, and a treasurer, with no person holding two offices.

(a) How many outcomes are possible?

(b) How many if a particular member must be president?

(c) How many if a particular member must hold *some* office?

Approved solution

(a) $P(10, 3) = 10 \cdot 9 \cdot 8 = 720$.
(b) President fixed; the other two offices from the remaining 9: $P(9, 2) = 72$.
(c) Choose which of the 3 offices that member holds (3 ways), then fill the other two from the remaining 9: $3 \cdot P(9, 2) = 3 \cdot 72 = 216$.
Alternatively, by complement: $720 - P(9, 3) = 720 - 504 = 216$. ✓

Two routes, same answer — always a good check when a restriction is awkward.

Section C — Writing

5. In four to six complete sentences, explain why $P(n,r) = \dfrac{n!}{(n-r)!}$ follows from the Multiplication Principle. Say what $(n-r)!$ represents.

Approved solution

Model answer. Building an ordered list of r items from n is a sequence of r stages: the first position has n choices, the second $n-1$ since one item is used, and so on down to $n-r+1$ choices for the rth. By the Multiplication Principle the total is the product $n(n-1)\cdots(n-r+1)$. That product is the beginning of $n!$, stopping early. To write it as a quotient, note that $n!$ continues with the factors $(n-r)(n-r-1)\cdots 1$, which is exactly $(n-r)!$, so dividing $n!$ by $(n-r)!$ cancels the factors we did not use. Thus $(n-r)!$ represents the arrangements of the $n-r$ items left unchosen, which the formula removes.

Full credit requires saying what $(n-r)!$ cancels. An answer that only restates the formula earns about half.

Section D — Challenge (optional)

6. In how many ways can 5 people be seated around a *circular* table, if two seatings are the same when one is a rotation of the other?

Approved solution

In a row there are $5! = 120$ arrangements. Around a circle, each arrangement can be rotated into 5 positions that all count as the same seating. So divide:

$$\frac{5!}{5} = 4! = 24.$$

This is the divide-by-the-over-count technique that Wednesday will use to build combinations. Here we over-counted by a factor of 5 — the rotations — and corrected for it.
In general, n people around a circular table give $(n-1)!$ seatings.

Instructor notes

- **Problem 2's gluing trick** recurs constantly. Teach it as a named move.
- **Problem 4(c)** has two routes — direct and complement — and agreement between them is a check worth modelling.
- **Problem 6 previews Wednesday**: divide by the over-count. Assign it if you want combinations to land faster.
- **Problem 3** reinforces handling the restricted position first.
- Reserved for Homework 13: Section 8.4 even-numbered exercises. Odd ones have printed answers and are self-check only.